Appendix: Exclusions

Population or set	Proteins	Relation
AFShieldingSet	625	AlphaFold/ORCA shielding set: 521 fitting, 25 selector and 79 protected proteins.
RefDB/OpenFold3 evaluated set	1,915	RefDbSmallSet (173) plus RefDbTestLargeSet (1,742).
RefDbSmallSet	173	One member list with static and GROMACS preparations.
RefDbTestLargeSet	1,742	External static comparison; no fitting or model selection.
RefDbTestSmallSet	386	Ligand- and bound-metal-free subset of RefDbTestLargeSet; not an additional population.

Table 1: Protein populations used by the reported models and exclusion counts.

Context	n	Entries	Reason not evaluated
External static application	4	bmr4443, bmr4967, bmr5342, bmr6389	The fitted static model had no target category for TRP:H1, TRP:H2 or TRP:H3.

Table 2: Four external entries not evaluated by the fitted static model. They are not part of the 1,915-member evaluated set or any target-row percentage.

Preparation	n	Entries	Reason
GROMACS	9	bmr4281, bmr4792, bmr5798, bmr5867, bmr5879, bmr6015, bmr6221, bmr6396, bmr6601	Technical issue.

Table 3: GROMACS preparation exclusions outside the final 173-member RefDbSmallGromacsSet. These are separate from the four external static exclusions.

Excluded AFShieldingSet target	Rows	Basis
H1	616	Terminal preparation
H2	625	Terminal preparation
H3	625	Terminal preparation
OXT	625	Terminal preparation
CYS:SG	489	Cysteine-sulfur preparation

Table 4: Shielding target exclusions across all 625 AFShieldingSet proteins: 413,563 source rows, 410,583 retained targets and 2,980 excluded targets. Excluded atoms remained graph nodes but did not enter the shielding target loss.

Population or preparation	Proteins	Source rows	Retained	Omitted	Omitted (%)
RefDB/OpenFold3 evaluated set	1,915	1,612,984	1,608,401	4,583	0.284%
RefDbSmallTleapSet	173	143,953	143,688	265	0.184%
RefDbTestLargeSet	1,742	1,469,031	1,464,713	4,318	0.294%
RefDbSmallGromacsSet	173	143,953	143,691	262	0.182%

Table 5: RefDB shift-row retention. RefDbSmallTleapSet and RefDbSmallGromacsSet are two preparations of the same 173 proteins and are not added together. RefDbTestLargeSet is the external part of the 1,915-member evaluated set.

Reason	Rows	Source rows (%)
Ambiguous assignment did not define a complete set of possible atoms.	3,211	0.1991%
Several source assignments mapped to one prepared atom.	553	0.0343%
No matching deposited assignment was established.	298	0.0185%
RefDB correction did not reconcile with the deposited value.	74	0.0046%
Invalid source metadata.	29	0.0018%
Named atom was absent from the prepared structure.	27	0.0017%
Duplicate corrected assignment identity.	18	0.0011%
N-terminal H could not be matched to prepared H1/H2/H3.	193	0.0120%
HIE preparation has no HIS HD1.	123	0.0076%
ASP preparation has no HD2.	3	0.0002%
Disulfide CYS preparation has no HG.	2	0.0001%
VAL/LEU branch assignment lacked its symmetry partner.	52	0.0032%
Total	4,583	0.2841%

Table 6: Reasons for omitted RefDB shift rows in the 1,915-member evaluated OpenFold3 set.

Nucleus	Source rows	Retained	Omitted	All source rows (%)
H	875,724	872,433	3,291	0.2040%
C	562,182	561,025	1,157	0.0717%
N	175,078	174,943	135	0.0084%
Total	1,612,984	1,608,401	4,583	0.2841%

Table 7: Retained and omitted RefDB shift rows by nucleus in the 1,915-member evaluated OpenFold3 set.

Entry	Static	GROMACS	Entry	Static	GROMACS	Entry	Static	GROMACS	Entry	Static	GROMACS
bmr260	2	—	bmr274	2	—	bmr294	6	—	bmr295	6	—
bmr443	2	—	bmr555	2	—	bmr979	8	—	bmr1061	4	—
bmr1062	4	—	bmr1162	1	—	bmr1517	2	—	bmr1658	2	—
bmr1673	2	—	bmr1918	1	—	bmr2065	2	—	bmr2281	1	—
bmr2410	4	—	bmr2868	1	—	bmr3441	1	—	bmr3442	2	—
bmr4012	46	—	bmr4022	1	—	bmr4033	1	—	bmr4035	8	—
bmr4036	6	—	bmr4037	3	—	bmr4039	4	—	bmr4041	5	—
bmr4043	6	—	bmr4044	1	—	bmr4045	20	—	bmr4047	12	—
bmr4049	14	—	bmr4054	4	—	bmr4055	10	—	bmr4060	8	—
bmr4069	1	—	bmr4070	2	—	bmr4072	3	—	bmr4076	20	—
bmr4078	1	—	bmr4081	6	—	bmr4082	6	6	bmr4083	8	—
bmr4085	6	—	bmr4087	10	—	bmr4094	2	—	bmr4095	23	23
bmr4098	9	—	bmr4110	3	—	bmr4114	2	—	bmr4117	3	—
bmr4121	10	10	bmr4127	28	—	bmr4131	23	—	bmr4133	1	—
bmr4136	2	—	bmr4143	10	—	bmr4149	1	—	bmr4156	6	—
bmr4157	11	—	bmr4158	8	—	bmr4159	1	—	bmr4161	6	—
bmr4162	2	—	bmr4163	1	—	bmr4168	1	—	bmr4173	3	—
bmr4184	5	—	bmr4193	7	—	bmr4195	1	—	bmr4197	5	—
bmr4202	5	—	bmr4204	5	—	bmr4206	29	—	bmr4207	5	—
bmr4210	25	—	bmr4211	4	—	bmr4216	6	—	bmr4219	2	—
bmr4224	4	—	bmr4236	3	—	bmr4237	3	3	bmr4241	1	—
bmr4242	8	—	bmr4254	2	2	bmr4255	4	—	bmr4257	5	—
bmr4261	3	—	bmr4278	4	—	bmr4280	1	—	bmr4290	1	—
bmr4295	2	—	bmr4297	1	—	bmr4299	30	—	bmr4300	29	—
bmr4301	1	1	bmr4305	6	—	bmr4308	11	—	bmr4309	1	—
bmr4318	13	—	bmr4327	4	—	bmr4333	3	—	bmr4334	4	—
bmr4335	10	—	bmr4336	20	—	bmr4340	26	—	bmr4343	1	—
bmr4348	1	1	bmr4350	1	—	bmr4360	3	—	bmr4370	20	—
bmr4378	20	—	bmr4380	1	—	bmr4393	10	—	bmr4394	11	—
bmr4417	2	—	bmr4421	8	8	bmr4422	1	—	bmr4425	1	—
bmr4426	1	—	bmr4427	5	—	bmr4429	1	—	bmr4431	14	—
bmr4437	29	—	bmr4449	10	—	bmr4452	1	—	bmr4458	8	—
bmr4459	8	—	bmr4460	1	—	bmr4466	17	—	bmr4467	1	—
bmr4473	2	—	bmr4475	10	—	bmr4486	2	—	bmr4494	11	—
bmr4553	33	—	bmr4558	21	—	bmr4568	1	—	bmr4579	3	—
bmr4581	3	3	bmr4587	8	—	bmr4590	20	20	bmr4591	1	—
bmr4615	4	—	bmr4619	2	—	bmr4628	1	—	bmr4637	20	—
bmr4638	1	—	bmr4639	1	—	bmr4642	2	—	bmr4644	2	—
bmr4661	1	—	bmr4666	3	—	bmr4674	41	—	bmr4677	3	—
bmr4688	17	—	bmr4696	4	—	bmr4712	25	—	bmr4714	2	—
bmr4716	19	—	bmr4727	29	—	bmr4740	4	—	bmr4760	4	—
bmr4782	2	—	bmr4786	34	—	bmr4789	16	—	bmr4798	1	—
bmr4814	6	—	bmr4819	1	—	bmr4829	2	—	bmr4834	2	—
bmr4836	4	—	bmr4839	27	—	bmr4840	3	—	bmr4857	5	—
bmr4863	1	1	bmr4874	1	—	bmr4878	1	—	bmr4884	6	—
bmr4886	17	—	bmr4893	2	—	bmr4896	1	—	bmr4899	2	—
bmr4900	2	—	bmr4908	36	—	bmr4909	24	—	bmr4911	1	—
bmr4912	3	—	bmr4918	9	—	bmr4924	3	—	bmr4929	4	—
bmr4931	4	—	bmr4932	4	—	bmr4934	3	—	bmr4957	18	—
bmr4961	3	—	bmr4964	6	—	bmr4965	8	—	bmr4969	19	—
bmr4971	2	—	bmr4978	12	—	bmr4981	2	2	bmr4984	9	—
bmr4985	12	—	bmr4991	9	—	bmr5000	1	—	bmr5006	6	—
bmr5022	1	—	bmr5028	2	—	bmr5030	2	—	bmr5039	1	—
bmr5051	1	—	bmr5058	6	—	bmr5059	5	—	bmr5060	11	—
bmr5067	2	—	bmr5069	1	—	bmr5070	10	—	bmr5077	7	—
bmr5082	5	—	bmr5083	17	—	bmr5093	5	—	bmr5097	2	—
bmr5099	16	—	bmr5104	1	—	bmr5126	1	—	bmr5128	24	—
bmr5129	12	—	bmr5143	1	—	bmr5149	1	—	bmr5175	1	—
bmr5176	1	—	bmr5182	1	—	bmr5194	16	—	bmr5210	2	—
bmr5217	10	—	bmr5218	1	—	bmr5232	1	—	bmr5242	2	—
bmr5257	1	—	bmr5261	11	—	bmr5264	9	—	bmr5267	5	—
bmr5275	1	—	bmr5284	1	—	bmr5285	1	—	bmr5311	3	3
bmr5312	4	—	bmr5313	8	—	bmr5317	19	—	bmr5320	14	—
bmr5322	1	—	bmr5326	17	—	bmr5328	1	1	bmr5329	3	3
bmr5347	36	—	bmr5354	18	—	bmr5356	7	—	bmr5364	1	—
bmr5375	1	1	bmr5393	11	—	bmr5398	20	—	bmr5402	33	—
bmr5408	1	—	bmr5410	1	—	bmr5454	3	—	bmr5455	9	—
bmr5457	2	—	bmr5464	1	—	bmr5473	24	—	bmr5475	1	—
bmr5485	10	—	bmr5492	3	—	bmr5495	1	—	bmr5499	1	1
bmr5512	1	—	bmr5516	7	—	bmr5540	12	—	bmr5555	38	—
bmr5563	4	4	bmr5565	6	—	bmr5571	29	—	bmr5572	1	—
bmr5583	1	—	bmr5589	1	—	bmr5594	2	—	bmr5595	9	—
bmr5606	1	—	bmr5609	1	—	bmr5610	8	—	bmr5621	1	—
bmr5629	8	—	bmr5662	1	—	bmr5665	6	—	bmr5668	21	—
bmr5670	2	—	bmr5677	10	—	bmr5689	15	15	bmr5701	12	—
bmr5706	1	—	bmr5710	11	—	bmr5740	19	—	bmr5741	36	—
bmr5744	1	—	bmr5749	2	—	bmr5754	1	—	bmr5767	2	—
bmr5778	1	—	bmr5779	2	—	bmr5783	1	—	bmr5789	4	—
bmr5801	2	2	bmr5806	1	—	bmr5808	1	—	bmr5810	1	—
bmr5817	8	—	bmr5843	1	—	bmr5844	1	—	bmr5846	1	0
bmr5866	2	—	bmr5873	11	11	bmr5875	2	2	bmr5880	12	—
bmr5884	24	—	bmr5891	19	—	bmr5898	3	—	bmr5905	1	—
bmr5908	1	—	bmr5909	1	—	bmr5910	1	—	bmr5912	4	—
bmr5924	2	—	bmr5933	16	—	bmr5935	2	—	bmr5940	8	—
bmr5948	1	—	bmr5949	1	—	bmr5952	5	—	bmr5958	1	—
bmr5961	2	—	bmr5963	1	1	bmr5974	2	—	bmr5998	25	—

continued

Entry	Static	GROMACS	Entry	Static	GROMACS	Entry	Static	GROMACS	Entry	Static	GROMACS
bmr5999	3	—	bmr6002	22	—	bmr6012	24	—	bmr6022	7	—
bmr6028	1	1	bmr6029	3	—	bmr6031	1	1	bmr6032	1	—
bmr6044	3	—	bmr6046	13	—	bmr6051	2	—	bmr6060	18	—
bmr6070	4	—	bmr6071	16	—	bmr6072	6	—	bmr6080	12	—
bmr6082	20	—	bmr6089	7	—	bmr6091	5	—	bmr6092	11	—
bmr6096	18	—	bmr6110	3	—	bmr6121	29	—	bmr6122	3	3
bmr6123	2	—	bmr6127	21	—	bmr6132	42	—	bmr6133	17	—
bmr6135	8	—	bmr6137	12	—	bmr6150	19	—	bmr6159	9	—
bmr6160	10	—	bmr6176	1	—	bmr6177	9	—	bmr6178	5	—
bmr6184	3	—	bmr6185	12	—	bmr6188	2	2	bmr6198	5	—
bmr6199	1	—	bmr6209	2	2	bmr6222	1	—	bmr6225	4	—
bmr6231	7	—	bmr6233	1	—	bmr6238	4	—	bmr6241	1	—
bmr6252	1	—	bmr6254	1	1	bmr6255	12	—	bmr6256	1	—
bmr6257	1	—	bmr6264	2	—	bmr6268	17	—	bmr6271	1	—
bmr6279	16	—	bmr6285	2	—	bmr6292	2	—	bmr6298	1	—
bmr6299	18	—	bmr6300	2	—	bmr6317	28	—	bmr6321	1	—
bmr6322	1	—	bmr6334	3	—	bmr6338	11	—	bmr6339	22	—
bmr6344	3	—	bmr6349	1	—	bmr6354	10	—	bmr6355	6	6
bmr6356	3	—	bmr6362	12	—	bmr6371	3	—	bmr6375	11	11
bmr6376	2	2	bmr6380	1	1	bmr6400	2	—	bmr6401	1	—
bmr6406	10	—	bmr6407	1	—	bmr6410	7	—	bmr6447	2	—
bmr6448	2	—	bmr6455	12	—	bmr6460	23	—	bmr6465	1	—
bmr6466	4	4	bmr6469	1	1	bmr6476	2	—	bmr6493	13	—
bmr6498	9	—	bmr6500	1	—	bmr6501	5	—	bmr6503	4	—
bmr6520	3	—	bmr6525	3	—	bmr6527	49	49	bmr6536	1	—
bmr6538	1	—	bmr6540	1	—	bmr6554	5	—	bmr6555	1	—
bmr6563	6	—	bmr6564	4	—	bmr6571	5	—	bmr6575	3	—
bmr6578	9	—	bmr6579	6	—	bmr6580	6	—	bmr6585	27	—
bmr6587	1	—	bmr6588	1	—	bmr6589	1	—	bmr6590	3	—
bmr6592	1	—	bmr6596	1	—	bmr6597	31	—	bmr6598	1	—
bmr6600	10	—	bmr6612	2	—	bmr6621	4	—	bmr6629	1	—
bmr6635	1	—	bmr6639	3	—	bmr6647	4	—	bmr6648	1	—
bmr6649	1	1	bmr6654	9	—	bmr6688	51	—	bmr6717	8	—
bmr6718	7	7	bmr6724	1	—	bmr6738	14	—	bmr6739	23	—
bmr6746	1	—	bmr6747	1	—	bmr6754	1	—	bmr6759	1	—
bmr6761	1	—	bmr6769	1	—	bmr6771	1	—	bmr6776	5	—
bmr6779	1	—	bmr6787	4	—	bmr6807	1	—	bmr6820	3	—
bmr6821	3	—	bmr6824	2	—	bmr6843	18	—	bmr6850	11	—
bmr6851	2	—	bmr6866	5	—	bmr6872	1	—	bmr6882	39	—
bmr6893	2	—	bmr6894	2	—	bmr6899	20	—	bmr6902	21	—
bmr6918	6	—	bmr6920	1	—	bmr6928	19	—	bmr6932	5	5
bmr6969	6	—	bmr6970	1	—	bmr6973	1	—	bmr6980	30	—
bmr6981	11	11	bmr6989	1	1	bmr7020	2	—	bmr7032	3	—
bmr7057	8	8	bmr7067	3	—	bmr7069	1	—	bmr7079	1	—
bmr7083	6	—	bmr7087	45	—	bmr7089	1	—	bmr7092	6	—
bmr7094	1	1	bmr7103	26	—	bmr7104	4	4	bmr7106	2	—
bmr7109	10	10	bmr7112	9	—	bmr7115	3	—	bmr7129	47	—
bmr7136	2	—	bmr7159	2	—	bmr7160	2	—	bmr7172	1	—
bmr7174	1	—	bmr7201	5	—	bmr7210	4	—	bmr7211	3	—
bmr7220	10	—	bmr7241	1	—	bmr7251	1	—	bmr7259	1	—
bmr7264	7	—	bmr7310	1	—	bmr7314	1	—	bmr7315	1	—
bmr7324	2	—	bmr7357	1	—	bmr7371	2	2	bmr7399	1	—
bmr7400	3	—	bmr10004	42	—	bmr10015	2	—	bmr10027	1	—
bmr10043	1	—	bmr10065	2	—	bmr10118	1	1	bmr11062	2	—
bmr15001	23	—	bmr15002	1	—	bmr15014	12	—	bmr15037	4	—
bmr15074	7	—	bmr15085	1	—	bmr15129	1	1	bmr15161	1	—
bmr15204	1	—	bmr15232	2	—	bmr15242	1	—	bmr15332	1	0
bmr15333	1	—	bmr15355	1	0	bmr15372	6	—	bmr15406	15	—
bmr15412	2	—	bmr15420	32	—	bmr15421	7	—	bmr15424	15	—
bmr15429	45	—	bmr15498	17	—	bmr15503	4	—	bmr15507	2	—
bmr15508	1	—	bmr15548	1	—	bmr15560	4	—	bmr15564	1	—
bmr15669	1	—	bmr15756	2	2	bmr15757	1	—	bmr15783	3	—
bmr15785	10	—	bmr15792	1	—	bmr15800	1	—	bmr15826	2	—
bmr15928	37	—	bmr15952	1	—	bmr16024	1	—	bmr16074	1	—
bmr16169	2	—	bmr16171	4	—	bmr16235	1	—	bmr16294	8	—
bmr16310	7	—	bmr16334	11	—	bmr16362	1	—	bmr16367	1	—
bmr16379	32	—	bmr16447	1	—	bmr16466	20	—	bmr16493	2	—
bmr16494	2	—	bmr16502	2	—	bmr16523	16	—	bmr16524	34	—
bmr16525	34	—	bmr16526	33	—	bmr16527	32	—	bmr16528	34	—
bmr16541	1	—	bmr16587	1	—	bmr16668	1	—	bmr16670	2	—
bmr16847	1	—	bmr16878	14	—	bmr16904	13	—	bmr16925	2	—
bmr17008	3	—	bmr17357	4	—	bmr17393	23	—	bmr17415	1	—
bmr17476	1	—									

Table 8: Omitted RefDB shift rows for each affected entry. Static is the OpenFold3 preparation. GROMACS is shown only for members of RefDbSmallSet; an em dash means that the external protein had no GROMACS preparation.